6. Create Address class with data members

HouseNo

Colony

Area

City

Pincode

Implement followings

- **Default constructor**
- **Parameterized constructor**
- **Accept Function for data input**
- **Display function for data display**
- **Separate header and implementation files**
- **Write function to find whether two addresses are same or different.**
- **Overlaod == operator.**

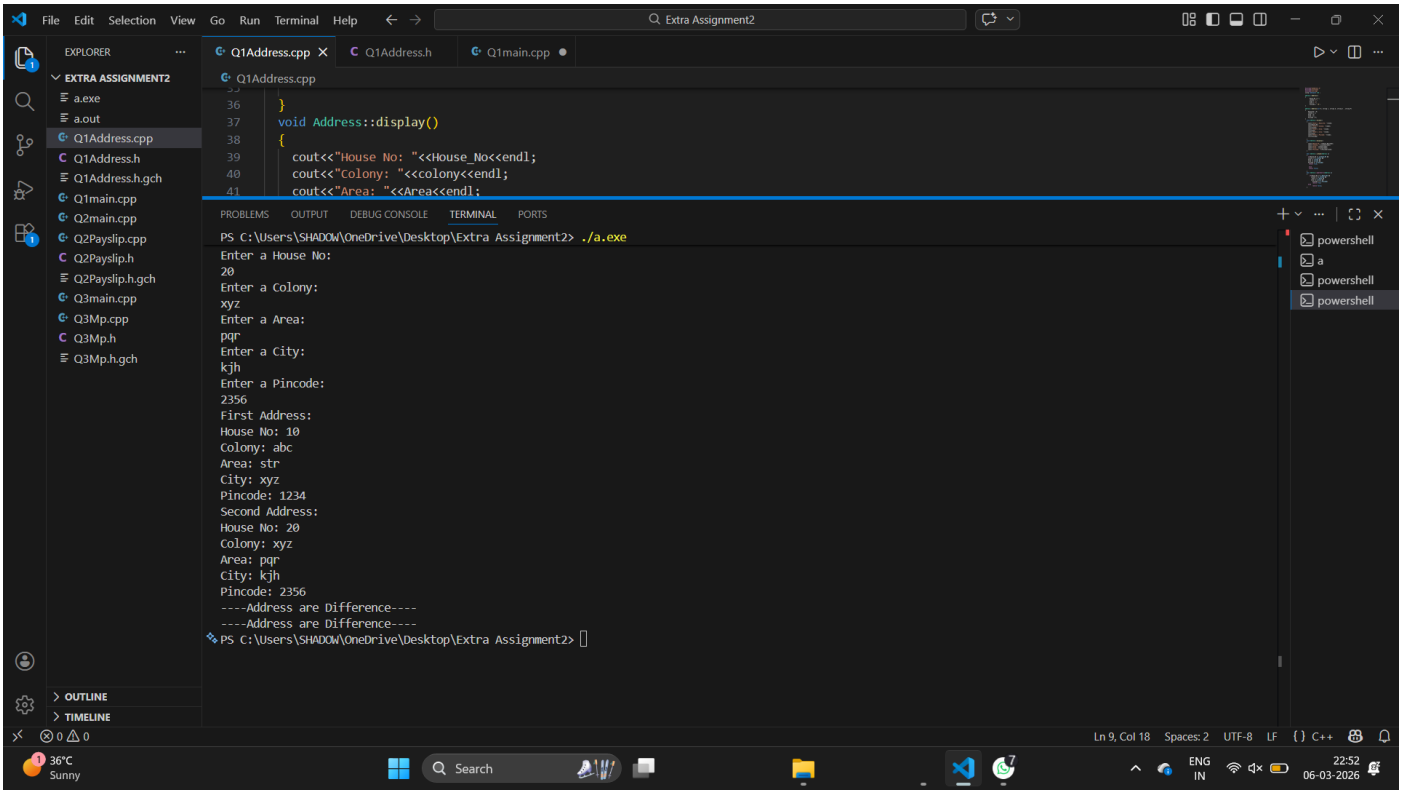
```
#ifndef Q1ADDRESS_H
#define Q1ADDRESS_H

#include<iostream>
using namespace std;

class Address{
  private:
  int House_No ;
  string colony;
  string Area;
  string City;
  string Pincode;

  public:
  Address();

  Address(int , string , string , string , string );

  void Accept();
```

```
   void display();

   bool isSame(Address);

   bool operator==(Address);

};
#endif
#include"Q1Address.h"
#include<iostream>
#include<cstring>
using namespace std ;

Address::Address()
{
    House_No = 0 ;
    colony ="";
    Area = "";
    City = "";
    Pincode = '\0';
}

Address::Address(int H, string c, string A, string C , string P)
{
   House_No = H;
   colony = c;
   Area = A;
   City = C;
   Pincode = P;
}
 void Address::Accept()
  {
   cout<<"Enter a House No: "<<endl;
```

```
  cin>>House_No;
  cout<<"Enter a Colony: "<<endl;
  cin>>colony;
  cout<<"Enter a Area: "<<endl;
  cin>>Area;
  cout<<"Enter a City: "<<endl;
  cin>>City;
  cout<<"Enter a Pincode: "<<endl;
  cin>>Pincode;

}
void Address::display()
{
  cout<<"House No: "<<House_No<<endl;
  cout<<"Colony: "<<colony<<endl;
  cout<<"Area: "<<Area<<endl;
  cout<<"City: "<<City<<endl;
  cout<<"Pincode: "<<Pincode<<endl;
}

bool Address::isSame(Address a)
{
  if(House_No == a.House_No &&
  colony == a.colony &&
  Area == a.Area &&
  City == a.City &&
  Pincode == a.Pincode)
   return true;

   else
   return false;

}
bool Address::operator==(Address a)
```

```
 {
    if(House_No == a.House_No &&
      colony == a.colony &&
       Area == a.Area &&
      City == a.City &&
       Pincode == a.Pincode)
       return true;
  else
       return false;
 }
#include<iostream>
#include<cstring>
#include "Q1Address.h"

using namespace std ;
 int main ()
 {
   Address a1 , a2;

   cout<<"Enter a First Address: "<<endl;
   a1.Accept();

   cout<<"Enter a Second Address: "<<endl;
   a2.Accept();

   cout<<"First Address: "<<endl;
   a1.display();

   cout<<"Second Address: "<<endl;
   a2.display();

   //using Function
   if (a1.isSame(a2))
      cout<<"----Address are Same----"<<endl;
```

```
    else
        cout<<"----Address are Difference----"<<endl;

    // Operator ==
    if( a1==a2)
        cout<<"----Address are Same----"<<endl;
    else
        cout<<"----Address are Difference----"<<endl;

    return 0;
}
```



7. Create Payslip class with data members

EmpNo

EmpName

DA

HRA

BasicSalary

Implement followings

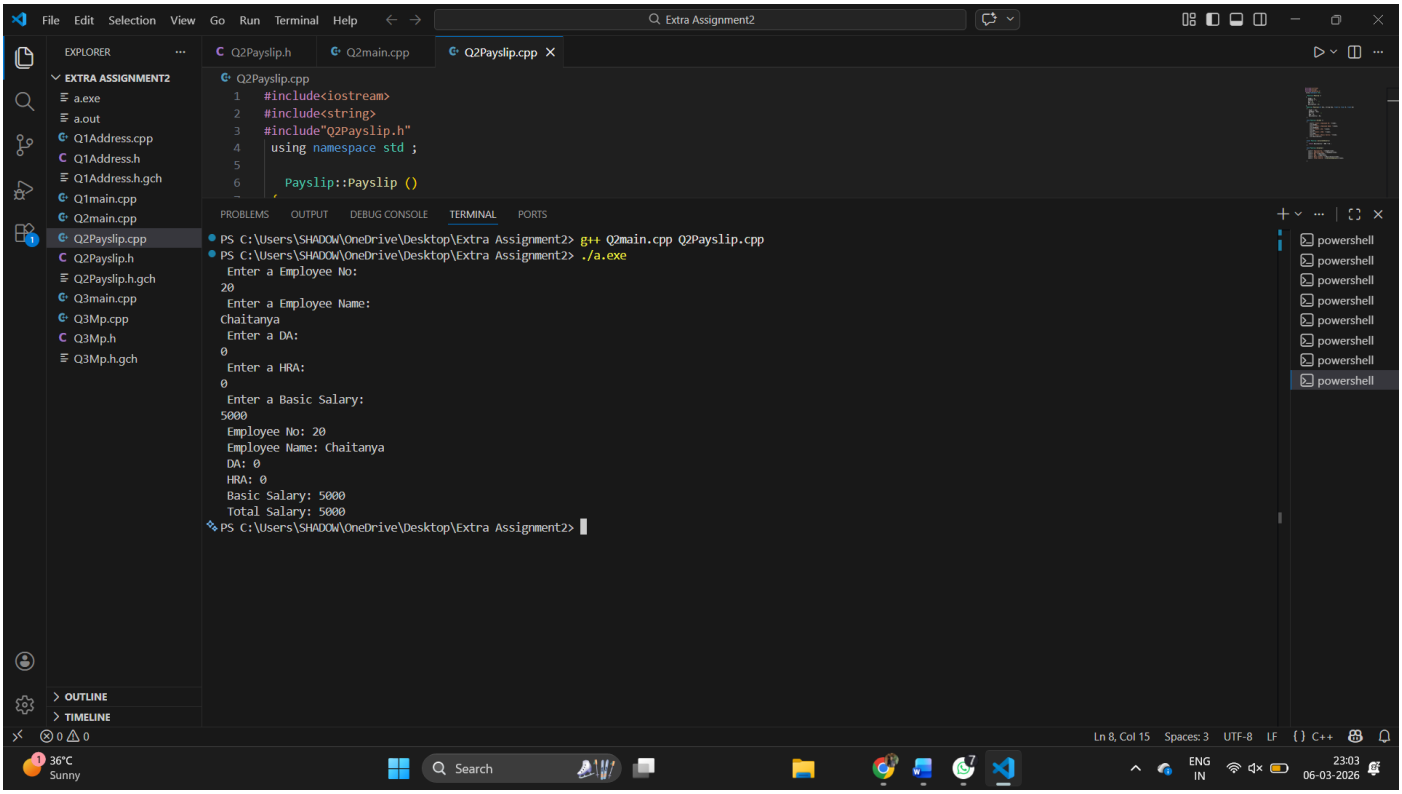
- **Default constructor**
- **Parameterized constructor**

- **Accept Function for data input**
- **Display function for data display**
- **Function to calculate salary from DA, HRA and Basic Salary**
- **Separate header and implementation files**

```
#ifndef Q1PAYSLIP_H
#define Q1PAYSLIP_H
#include<iostream>
using namespace std;
class Payslip
{
  private:
      int EmpNo;
      string EmpName;
      float DA;
      float HRA;
      float BasicSalary;

  public :

  Payslip(); //

  Payslip( int , string, float, float, float); //

  void Accept();

  void display();

  float calculateSalary();

};

#endif
#include<iostream>
#include<string>
```

```
#include"Q2Payslip.h"
 using namespace std ;


  Payslip::Payslip ()
{
   EmpNo = 0;
   EmpName ="";
   DA = 0;     //
   HRA = 0;
   BasicSalary = 0;
}
 Payslip::Payslip(int Eno, string Ena, float D, float H, float Bs)
 {
   EmpNo = Eno;
   EmpName = Ena;
   DA = D;       //
   HRA = H;
   BasicSalary = Bs;
 }

 void Payslip::Accept ()
 {
    cout<<" Enter a Employee No: "<<endl;
    cin>>EmpNo;
    cout<<" Enter a Employee Name: "<<endl;
    cin>>EmpName;
    cout<<" Enter a DA: "<<endl;
    cin>>DA;
    cout<<" Enter a HRA: "<<endl;
    cin>>HRA;
    cout<<" Enter a Basic Salary: "<<endl;
    cin>>BasicSalary;
 }
```

```
 float Payslip::calculateSalary()
 {
   return BasicSalary + HRA + DA ;
 }

 void Payslip::display()
 {
  cout<<" Employee No: "<<EmpNo<<endl;
  cout<<" Employee Name: "<<EmpName<<endl;
  cout<<" DA: "<<DA<<endl;
  cout<<" HRA: "<<HRA<<endl;
  cout<<" Basic Salary: "<<BasicSalary<<endl;
  cout<<" Total Salary: "<<calculateSalary()<<endl;

 }
#include<iostream>
#include<cstring>
#include"Q2Payslip.h"

using namespace std;
 int main ()
 {
     Payslip p;

     p.Accept();
     p.display();

     return 0;
 }
```



8. Create MobilePhone class with data members

ModelNo

Price

Manufacturer

Quantity

Implement followings

- **Default constructor**
- **Parameterized constructor**
- **Accept Function for data input**
- **Display function for data display**
- **Function to search MobilePhone model availability**
- **Write menu driven program with following menus**

o Add Mobile

o Display List of available mobiles

o Find out quantity of particular mobile type

o Find out availability of Particular model.

```
#ifndef Q2MP_H

#define Q2Mp_H

#include <iostream>

#include<cstring>
```

```
 using namespace std;

class MobilePhone
{
   private:
       int ModelNo;
       float Price;
       string Manufacturer;
       int Quantity;

   public:

      MobilePhone();

      MobilePhone(int, float , string, int);

      void Accept ();



      void display();

      int getModelNo();

      int getQuantity();

};
#endif
#include<iostream>
#include<cstring>
#include"Q3Mp.h"
using namespace std;

MobilePhone::MobilePhone()
{
```

```
    ModelNo = 0;
    Price = 0;
    Manufacturer = "";
    Quantity = 0;
}
 MobilePhone::MobilePhone(int M, float P, string Man, int Q)
{
    ModelNo = M;
    Price = P;
    Manufacturer = M;
    Quantity = Q;
}
void MobilePhone::Accept()
{
   cout<<" Enter a Model No: "<<endl;
    cin>>ModelNo;
    cout<<" Enter a Price: "<<endl;
    cin>>Price;
    cout<<" Enter a Manufacturer: "<<endl;
    cin>>Manufacturer;
    cout<<" Enter a Quantity: "<<endl;
    cin>>Quantity;
}
void MobilePhone::display()
{
   cout<<" Model No: "<<ModelNo<<endl;
   cout<<" Price: "<<Price<<endl;
   cout<<" Manufacturer: "<<Manufacturer<<endl;
   cout<<" Quantity: "<<Quantity<<endl;
}
int MobilePhone::getModelNo()
{
   return ModelNo;
}
```

```
int MobilePhone::getQuantity()
{
   return Quantity;
}
#include <iostream>
#include<cstring>
#include"Q3Mp.h"

using namespace std;
 int main ()
 {
   MobilePhone M[50];
   int count = 0;
   int choice ;
   int searchModel;

   do
   {
     cout<<"1.Mobile Shop Menu"<<endl;
     cout<<"2.Add Mobile"<<endl;
     cout<<"3.Display list of Mobiles"<<endl;
     cout<<"4.Find Quantity of Mobile"<<endl;
     cout<<"5.Check Availability of Model"<<endl;
     cout<<"6.Exit"<<endl;
     cout<<"7.Enter Choice"<<endl;
     cin>> choice;

     switch(choice)
     {
      case 1:
           M[count].Accept();
           count++;
           break;
```

```
    case 2:
        for(int i=0;i<count;i++)
        {
          M[i].display();
        }
        break;

    case 3:
        cout<<"Enter Model No to Find Quantity:";
        cin>>searchModel;
        for(int i=0; i<count; i++)
        {
          if(M[i].getModelNo()==searchModel)
          {
            cout<<"Model Available"<<endl;
          }

        }
         break;
    }
  }while(choice!=5);
  return 0;
}
```

File Edit Selection View Go Run Terminal Help

Extra Assignment2

EXPLORER

EXTRA ASSIGNMENT2

- a.exe
- a.out
- Q1Address.cpp
- Q1Address.h
- Q1Address.h.gch
- Q1main.cpp
- Q2main.cpp
- Q2Payslip.cpp
- Q2Payslip.h
- Q2Payslip.h.gch
- Q3main.cpp
- Q3Mp.cpp
- Q3Mp.h
- Q3Mp.h.gch

OUTLINE

TIMELINE

Q3Mp.h | Q3Mp.cpp | Q3main.cpp

Q3main.cpp

```cpp
#include <iostream>
#include<cstring>
#include"Q3Mp.h"

using namespace std;
  int main ()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\SHADOW\OneDrive\Desktop\Extra Assignment2> g++ Q3main.cpp Q3Mp.cpp
PS C:\Users\SHADOW\OneDrive\Desktop\Extra Assignment2> ./a.exe
1.Mobile Shop Menu
2.Add Mobile
3.Display list of Mobiles
4.Find Quantity of Mobile
5.Check Availability of Model
6.Exit
7.Enter Choice
1
 Enter a Model No:
5
 Enter a Price:
5000
 Enter a Manufacturer:
india
 Enter a Quantity:
2
1.Mobile Shop Menu
2.Add Mobile
3.Display list of Mobiles
4.Find Quantity of Mobile
5.Check Availability of Model
6.Exit
7.Enter Choice
2
 Model No: 5
 Price: 5000
 Manufacturer: india
 Quantity: 2
1.Mobile Shop Menu
```

powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
a

Ln 17, Col 48 Spaces: 4 UTF-8 LF { } C++

36°C Sunny

Search

ENG IN

23:07 06-03-2026